

UTKARSH SHARMA

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Research Interests: Differentiable & Neural Rendering, Physically-Based Rendering, Inverse Graphics

Education

Indian Institute of Science (IISc)

Bangalore, India

M.Tech in Computer Science and Engineering | **CGPA: 9.1/10.0**

2024 – 2026

Key Coursework: Graphics & Visualization, Learning for 3D Vision and Inverse Graphics, Deep Learning for Computer Vision, Computational Topology, Computational Methods of Optimization, Machine Learning for Signal Processing, Design and Analysis of Algorithms

Chandigarh University

Mohali, India

B.E. in Computer Science | **CGPA: 8.24/10.0**

2020 – 2024

Research Experience

M.Tech Thesis Project: Radiosity-Guided Accelerated Path Tracing | *CS 299, IISc*

Jan – Apr 2026

- Jointly advised by **Prof. Vijay Natarajan** (Visualization & Graphics Lab) and **Prof. R. Venkatesh Babu** (Vision & AI Lab).
- Proposed a camera-independent path guiding framework utilizing hardware-accelerated progressive radiosity to extract top- K radiant contributors to construct incoming radiance distribution.
- Designed a closed-form von Mises-Fisher (vMF) mixture model constructed on-the-fly, mapping precomputed radiometric geometry to continuous spherical distributions.
- Implemented a fused CUDA update kernel achieving bounded $\mathcal{O}(T)$ memory scaling and delivering up to $2.4\times$ higher sample throughput than NPM and PPG baselines on Diffuse dominant scenes with comparable variance reduction.
- **Radiosity-Guided Accelerated Path Tracing – Utkarsh Sharma**, Prashant Goswami, Rishubh Parihar, Vijay Natarajan, R. Venkatesh Babu. (*Under review at JCGT, July 2026*)

Selected Research & Projects

Futaba: Hardware-Accelerated GPU Path Tracer | *C++17, CUDA, OptiX, OpenGL*

Mar – Aug 2025

- Developed a GPU-accelerated path tracer in C++/CUDA with NVIDIA OptiX hardware ray tracing and an optimized software BVH fallback, tuning thread occupancy and GPU memory layouts.
- Implemented Next Event Estimation with multiple importance sampling, Russian Roulette, thin-lens depth of field, environment-map importance sampling, and Beckmann microfacet BSDFs (conductor/dielectric/plastic).
- Built an interactive OpenGL viewport (zero-copy PBO display) with multi-channel G-buffer visualizations (albedo, normals, depth, heatmap) and an OptiX AI denoiser for debugging and final output quality.

Neural Path Guiding for Path Tracing | *Mitsuba 3, PyTorch, Tiny-CUDA-NN*

Jun – Dec 2025

- Built a modular neural path-guiding framework in Mitsuba 3 (Python), implementing vMF mixture models, Neural Importance Sampling, and Distribution Factorization to guide ray directions, validating $2\text{--}4\times$ variance reduction in complex scenes.
- Developed a PyQt-based interactive GUI to inspect Mitsuba renders, visualize spherical probability density functions (PDFs), and debug neural network prediction behavior during rendering.
- Prototyped Neural Radiance Caching using online optimization and spatial hash-grid encodings.

Deep Learning-Based Denoising for Monte Carlo Rendering | *PyTorch, PBRT*

Jan – Apr 2025

- Evaluated and benchmarked diffusion-based and U-Net denoisers for low-sample Monte Carlo path tracing outputs in PyTorch and PBRT.
- Incorporated G-Buffer features (albedo, normals, depth, motion vectors) as multi-channel inputs to guide predictions, preserving geometric edges and material boundaries.
- Analyzed reconstruction quality using SSIM and PSNR metrics on PBRT-generated test scenes.

Neural Radiance Fields & 3D Gaussian Splatting | *PyTorch*

Jan – May 2025

- Built a NeRF pipeline from scratch in PyTorch, implementing the volumetric rendering equation, sinusoidal positional encoding, and hierarchical stratified sampling to reconstruct 3D scenes.
- Implemented 3D Gaussian Splatting, optimizing 3D covariance matrices and spherical harmonics (SH) coefficients; validated on the NeRF Synthetic dataset (PSNR > 25 dB).

Technical Writing

Introduction to Differentiable Rendering	July 2026
– Survey covering 6 seminal papers (Li 2018 through Zhang 2023), with step-by-step mathematical derivations, PyTorch implementations of Radiative and Path Replay Backpropagation.	
Introduction to 3D Rendering	Apr 2026
– Interactive graphics guide covering the rasterization pipeline, ray tracing and SDF sphere tracing implementations.	
Importance Sampling & Path Guiding for Path Tracing	Dec 2025
– Tutorial and survey spanning 9 papers (Vorba 2014 through Dahm 2025), covering Monte Carlo integration, MIS, normalizing flows, vMF mixtures, and neural path guiding.	
3D Reconstruction with 3DGS, NeRF, and Their Variants	Nov 2025
– Survey of 12 methods from NeRF to DUST3R/InstantSplat, with derivations of volume rendering, EWA splatting, hash encoding, and sparse-view reconstruction.	

Work Experience

Netradyne	Bangalore, India
<i>Machine Learning Engineer</i>	<i>July 2026 – Present</i>
– Analyze large-scale telematics and video-derived data to refine the GreenZone[®] Score , an AI-driven driver-performance metric, using Python, SQL, Pandas, and NumPy.	

Skills & Languages

Programming Languages: Python, C, C++ (17/20), CUDA, SQL, GLSL
Graphics & 3D Vision: Physically Based Rendering (PBR), Monte Carlo Path Tracing, Differentiable Rendering, Neural Rendering (NeRF, 3DGS), Volume Rendering, NVIDIA OptiX, OpenGL
Machine Learning & AI: PyTorch, Diffusion Models, U-Net, Neural Radiance Caching
Libraries & Tools: Mitsuba 3, PBRT, Tiny-CUDA-NN, Dr.Jit, Pandas, NumPy, Git
Spoken Languages: Hindi (Native), English (Fluent)

Teaching Experience

Teaching Assistant – Algorithms & Programming <i>Indian Institute of Science</i>	Aug – Dec 2025
Teaching Assistant – Computational Topology <i>Indian Institute of Science</i>	Jan – May 2026

Academic Achievements

Rank 8 out of 123,967 candidates in GATE Computer Science (CS) 2024 (<i>top 0.01%</i>)
Rank 17 out of 39,210 candidates in GATE Data Science & AI (DA) 2024 (<i>top 0.05%</i>)
Winner , CodeSmash 2.0 Competitive Programming Contest
Finalist , CodeRush 3.0 Competitive Programming Contest